

EXPERIMENT-17. Generate Contextual Embeddings

```
from transformers import BertTokenizer, BertModel
```

```
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
```

```
model = BertModel.from_pretrained("bert-base-uncased")
```

```
sentence = "Deep Learning is powerful."
```

```
inputs = tokenizer(sentence, return_tensors="pt")
```

```
outputs = model(**inputs)
```

```
print(outputs.last_hidden_state.shape)
```

output

```
Sentence: Artificial Intelligence is transforming the world.
Embedding Shape: torch.Size([1, 9, 768])
Contextual Embeddings:
tensor([[[[-0.0397,  0.1847,  0.0180, ..., -0.5428,  0.3945,  0.3015],
          [ 0.2583,  0.3813, -0.2127, ..., -0.2801,  0.5896,  0.5549],
          [ 0.1599,  0.4353, -0.1098, ..., -1.4470,  0.4459,  0.2455],
          ...,
          [ 0.7751, -0.3542,  0.3563, ..., -0.2726,  0.6736, -0.1412],
          [-0.1883, -0.6511, -0.1740, ...,  0.5566,  0.2397, -0.8522],
          [ 0.7499,  0.1488, -0.2976, ..., -0.1641, -0.4876, -0.2168]]]])
```

